

The Unit Circle and Exact Values

ANSWER KEY



Exact values

1 Write the exact value of each expression:

a $\sin \frac{\pi}{6} = \frac{1}{2}$

b $\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \frac{5\pi}{6} = -\frac{\sqrt{3}}{3}$

d $\sin \frac{3\pi}{2} = -1$

2 Write the exact value of each expression:

a $\cos \left(-\frac{\pi}{3}\right) = \frac{1}{2}$

b $\sin \frac{7\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \pi = 0$

d $\cos \frac{11\pi}{6} = \frac{\sqrt{3}}{2}$

Points on the unit circle

3 The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin \theta = \frac{4}{5}$

b $\cos \theta = -\frac{3}{5}$

c $\tan \theta = -\frac{4}{3}$

d the quadrant containing θ Quadrant II

4 State the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$

b $\frac{4\pi}{3} = \frac{\pi}{3}$

c $-\frac{\pi}{4} = \frac{\pi}{4}$

d $\frac{7\pi}{2} = \frac{\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

a $\sin \theta < 0$ and $\cos \theta > 0$ Quadrant IV.

b $\tan \theta > 0$ and $\sin \theta < 0$ Quadrant III.

6 Given that $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin \theta$ and $\tan \theta$ exactly.
 $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$, and in QIII $\sin \theta < 0$: $\sin \theta = -\frac{12}{13}$. $\tan \theta = \frac{-12/13}{-5/13} = \frac{12}{5}$.

Even/odd properties and periodicity

7 Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin \theta = 0$

b $\cos(-\theta) - \cos \theta = 0$

c $\frac{\sin(-\theta)}{\cos(-\theta)} = -\tan \theta$

8 Use periodicity to find the exact value of each expression:

a $\sin \frac{13\pi}{6} = \sin \frac{13\pi}{6} - 2\pi = \sin \frac{\pi}{6} = \frac{1}{2}$

b $\cos \frac{17\pi}{4} = \cos \frac{17\pi}{4} - 4\pi = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$

c $\tan \frac{9\pi}{4} = \tan \frac{9\pi}{4} - 2\pi = \tan \frac{\pi}{4} = 1$

Solving basic trigonometric equations on the unit circle

- 9 Find all solutions to $\sin \theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.
Reference angle $\frac{\pi}{6}$, sine negative in QIII and QIV: $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$.
- 10 Find all solutions to $2\cos \theta + \sqrt{3} = 0$ in the interval $[0, 2\pi)$.
 $\cos \theta = -\frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{6}$, cosine negative in QII and QIII: $\theta = \frac{5\pi}{6}, \frac{7\pi}{6}$.
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Building the unit circle from special triangles

- 11 Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.
A 30-60-90 triangle with hypotenuse 1 has legs $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$. At $\theta = \frac{\pi}{3}$ (60°), $x = \cos \frac{\pi}{3} = \frac{1}{2}$, $y = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$.
Point: $(\frac{1}{2}, \frac{\sqrt{3}}{2})$.
- 12 State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).
 $\theta = 0$: $(1, 0)$. $\theta = \frac{\pi}{2}$: $(0, 1)$. $\theta = \pi$: $(-1, 0)$. $\theta = \frac{3\pi}{2}$: $(0, -1)$.
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Using the Pythagorean identity with unit circle points

- 13 A point on the unit circle has x -coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y -coordinate using $x^2 + y^2 = 1$.
 $y^2 = 1 - \frac{49}{625} = \frac{576}{625}$, and in QIV $y < 0$: $y = -\frac{24}{25}$.
- 14 Given $\sin \theta = 0.6$ and θ in Quadrant II, find $\cos \theta$ and $\tan \theta$ exactly (using the fact $0.6 = \frac{3}{5}$).
 $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$, and in QII $\cos \theta < 0$: $\cos \theta = -\frac{4}{5}$. $\tan \theta = \frac{3/5}{-4/5} = -\frac{3}{4}$.
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The unit circle picture

- 15 The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos(\frac{2\pi}{3})$ and $\sin(\frac{2\pi}{3})$ visually, then state the exact values.
The angle $\frac{2\pi}{3}$ (120°) lies in QII, roughly two-thirds around from 0. Exact values: $\cos \frac{2\pi}{3} = -\frac{1}{2}$, $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$.
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All six trig functions from a unit circle point

- 16** A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ .

$$\sin \theta = \frac{12}{13}, \cos \theta = -\frac{5}{13}, \tan \theta = -\frac{12}{5}, \csc \theta = \frac{13}{12}, \sec \theta = -\frac{13}{5}, \cot \theta = -\frac{5}{12}.$$

Solving for the angle given exact coordinates

- 17** Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

This point lies in QIII where both coordinates are negative with reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ (only one solution in $[0, 2\pi)$ since the exact point pins down a single angle).